

Linear

Part A — Paràmetres de normalització (transforms)

Paràmetre	Valor
z_mean	-18.052
z_std	10.603
R_max	20
log_edep_mean	-4.4352
log_edep_std	2.9554
edep_offset_MeV	1e-12
log_t_mean	3.6413
log_t_std	1.8725
t_offset_ns	0.001
log_E_in_mean	-2.2675
log_E_in_std	3.1847
z_mean_poly_coeffs	[0.1187, 1.4921, 4.8059, -20.731]
condZ	✓

bins=8

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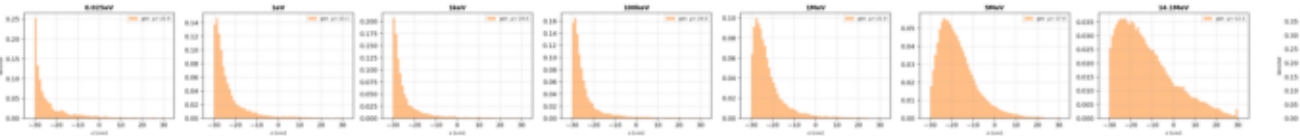
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C z phys

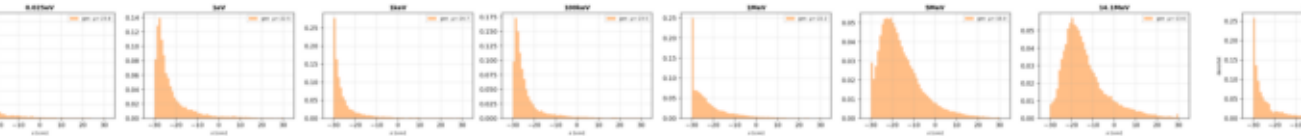
Linear

Part C — Distribution z Physa [cm] (truth vs generated)
Blue = truth (Dwarf4) Orange = generated (EPIC-FM)



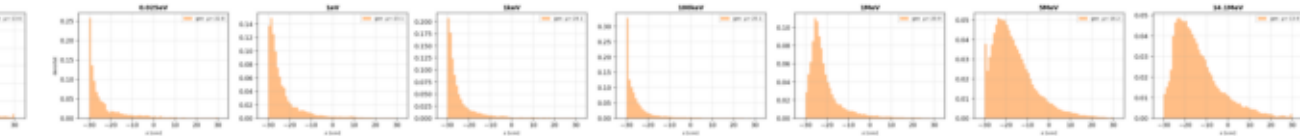
bins=8

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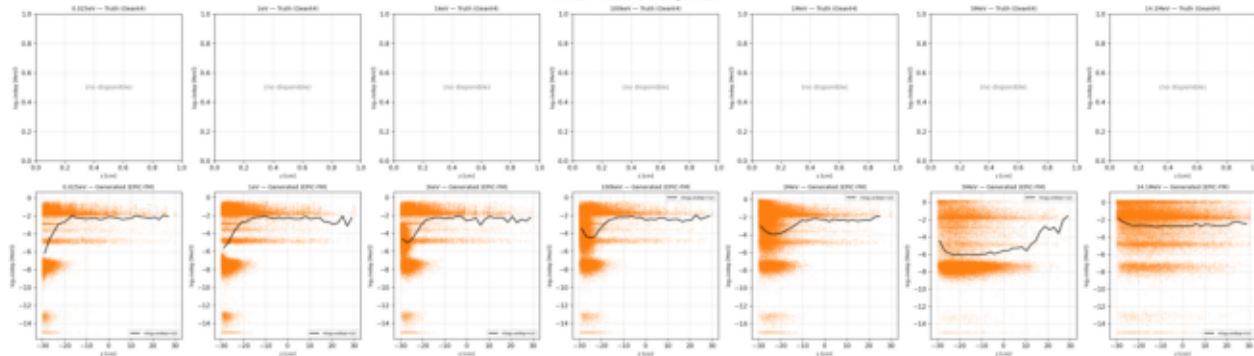
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D edep vs z scatter

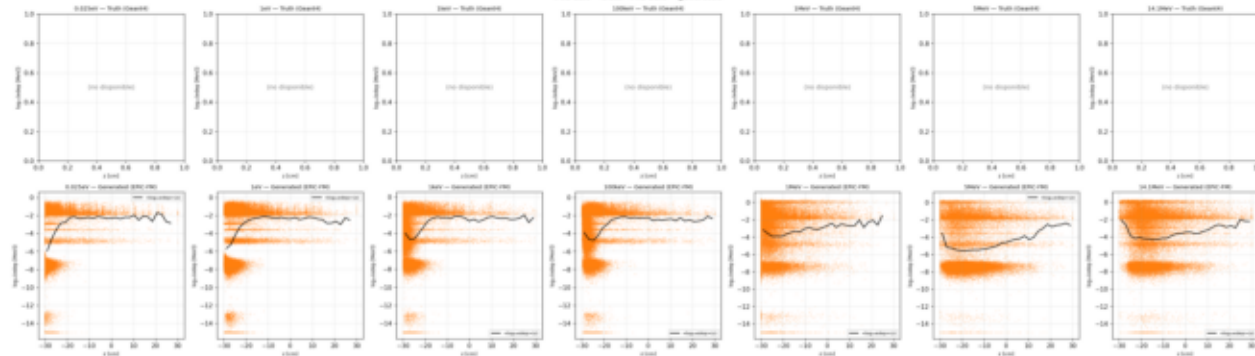
Linear

Part D -- Scatter edep vs z per hit
File data = truth File bins = generated



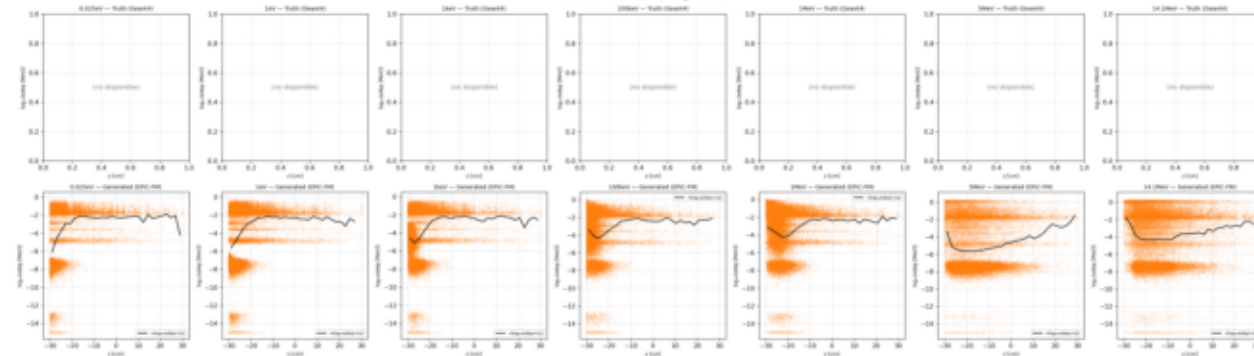
bins=8

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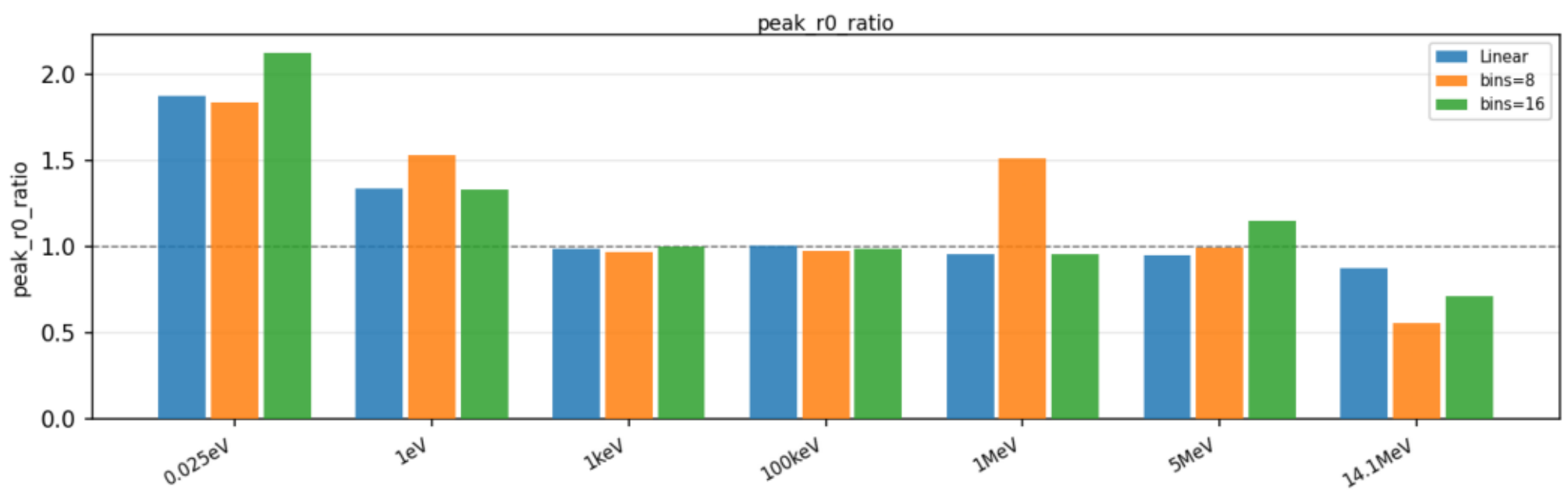
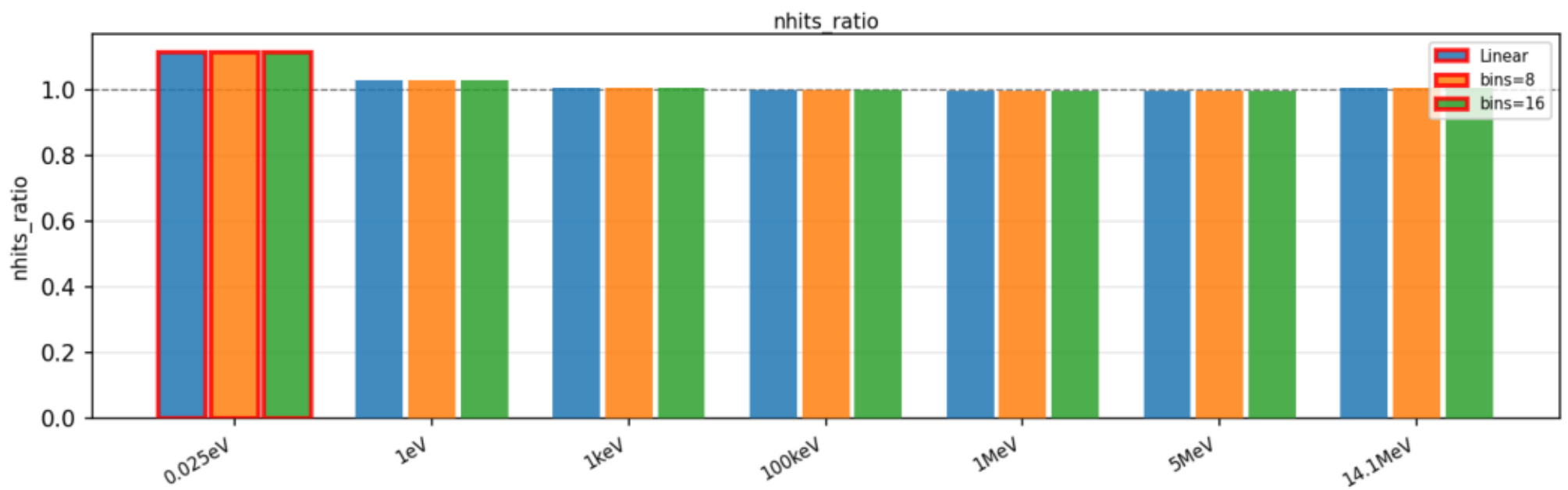
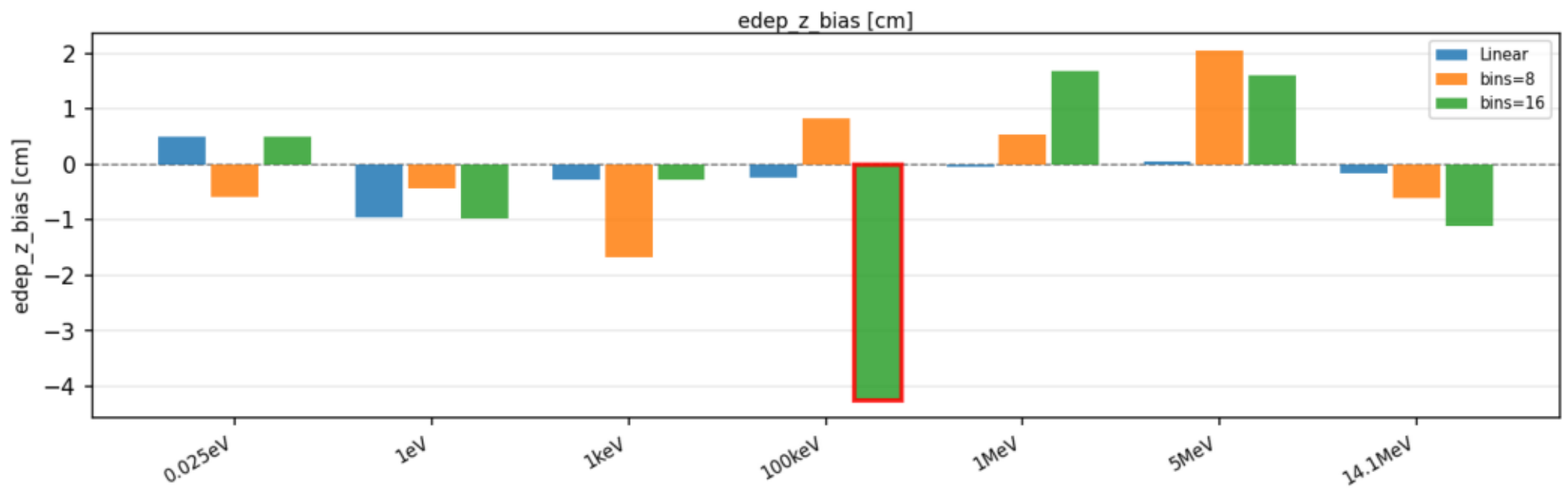
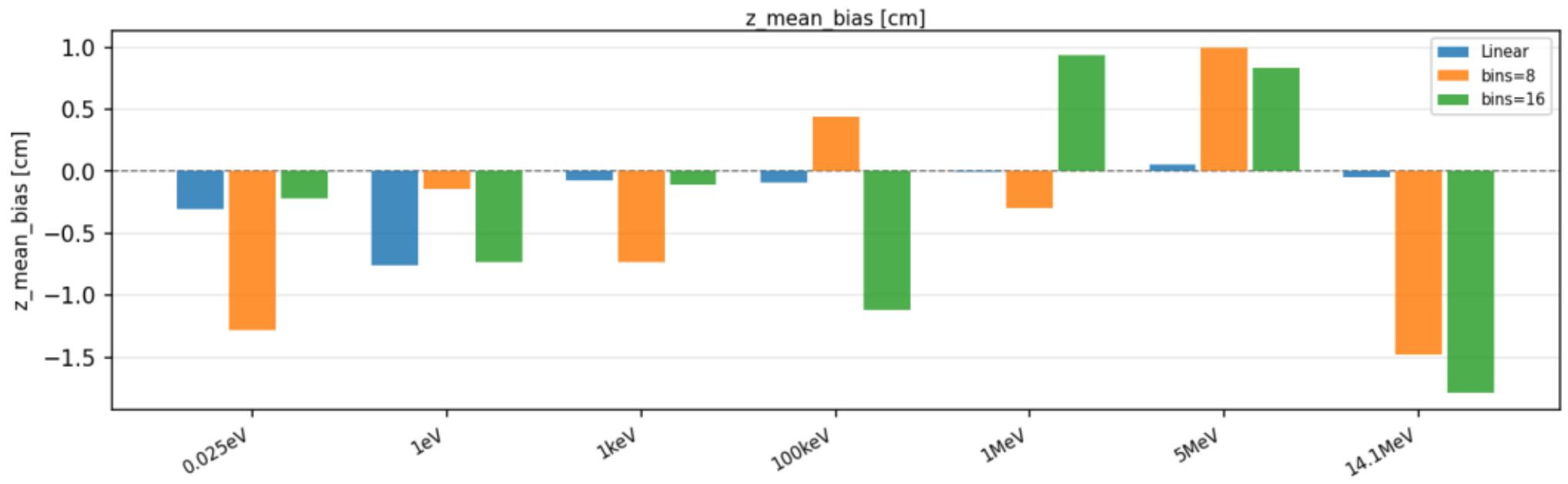
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Bandes horizontals esperades (neutrons en CH2):
 log10(edep) ~ +0.65 | 12C(n,n'g) inelastic, g = 4.44 MeV
 log10(edep) ~ +0.35 | H(n,g)D captura termica H, g = 2.22 MeV
 log10(edep) ~ +0.10 | 12C(n,g) captura C, g = 1.26 MeV
 continu | n(H,H)n recul de proto (elastic), E in [0, En]
 continu | Compton / fotoelectric de g secundaris
 log10(edep) ~ -14 | Floor de tracking Geant4 (electrons baixa E)

Part M - Resum metriques per run i energia



Part Z - Heatmap metriques OK/WRN/BAD per run i energia

